Machine Learning

Lecture 7: Why Regularization Works

Instructor: Prof. Udit Bhatia

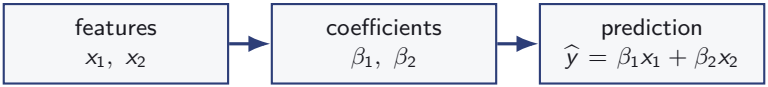
Civil Engineering; Computer Science and Engineering

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Start With OLS And Name Every Object

For each observation, two measured features x_1, x_2 produce one prediction:



Across all n observations, stack p features into $\boldsymbol{X} \in \mathbb{R}^{n \times p}$, the responses into $\boldsymbol{y} \in \mathbb{R}^n$, and the coefficients into $\boldsymbol{\beta} \in \mathbb{R}^p$. Then

$$\widehat{\boldsymbol{y}} = \boldsymbol{X}\boldsymbol{\beta}, \qquad J_{\mathrm{OLS}}(\boldsymbol{\beta}) = \frac{1}{2n}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2.$$

Summary

OLS chooses the coefficient vector $\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$ that makes the residual vector $\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}$ as small as possible on the training data.

Derive The Normal Equation Once, Then Reuse It

Expand the OLS objective before differentiating:

$$J_{\mathrm{OLS}}(\beta) = \frac{1}{2n}(\boldsymbol{y} - \boldsymbol{X}\beta)^\top(\boldsymbol{y} - \boldsymbol{X}\beta)$$
$$= \frac{1}{2n}\left(\boldsymbol{y}^\top\boldsymbol{y} - 2\beta^\top\boldsymbol{X}^\top\boldsymbol{y} + \beta^\top\boldsymbol{X}^\top\boldsymbol{X}\beta\right).$$

Differentiate and set the gradient to zero:

$$\nabla J_{\mathrm{OLS}}(\beta) = -\frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\beta),$$
$$0 = -\frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\widehat{\beta}),$$

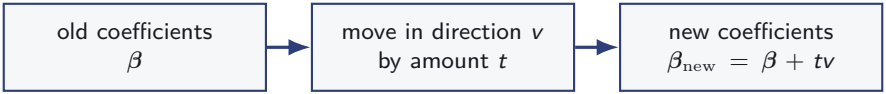
$$\boxed{\boldsymbol{X}^\top\boldsymbol{X}\widehat{\beta} = \boldsymbol{X}^\top\boldsymbol{y}.}$$

Summary

If $\hat{r} = \boldsymbol{y} - \boldsymbol{X}\widehat{\beta}$ is the fitted residual, the same equation says $\boldsymbol{X}^\top\hat{r} = 0$. This cancellation will matter when we study curvature.

What Changes When The Coefficients Move?

Start at any coefficient vector β. Choose a direction v in coefficient space and a step length t:



Now calculate the change in the fitted values, one algebraic step at a time:

$$\begin{aligned} \boldsymbol{X}\beta_{\text{new}} - \boldsymbol{X}\beta &= \boldsymbol{X}(\beta + tv) - \boldsymbol{X}\beta \\ &= t\boldsymbol{X}v. \end{aligned}$$

Insight

The vector $\boldsymbol{X}v$ is the prediction change caused by a one-unit coefficient move in direction v. This is the bridge from coefficients to the loss surface.

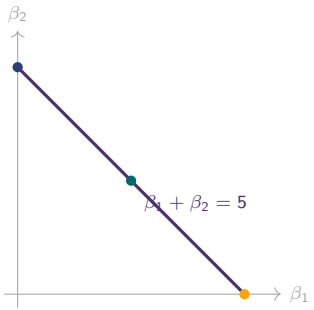
Exact Duplicate Features: The Prediction Only Sees Their Sum

Suppose two sensors measure exactly the same quantity: $x_2 = x_1$. Then, for every observation,

$$\widehat{y} = \beta_1 x_1 + \beta_2 x_2 = \beta_1 x_1 + \beta_2 x_1 = (\beta_1 + \beta_2) x_1.$$

What the data can identify

The data can identify the combined effect $\beta_1 + \beta_2$. They cannot decide how that total should be split between the two identical sensors.



Every point on the displayed line gives the same fitted values and the same OLS loss.

Three Very Different Coefficients Can Make The Same Prediction

Let $x_1 = x_2 = 1$ for one observation. All three coefficient pairs below predict the same value, even though they tell very different stories about the sensors.

$(\beta_1, \beta_2) = (2, 2)$
$2(1) + 2(1) = 4$

→

$(\beta_1, \beta_2) = (7, -3)$
$7(1) - 3(1) = 4$

→

$(\beta_1, \beta_2) = (100, -96)$
$100(1) - 96(1) = 4$

Start at $(2, 2)$ and move in the coefficient direction

$$v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Taking $t = 5$ gives $(2, 2) + 5v = (7, -3)$. One coefficient rises while the other falls by the same amount, so $\beta_1 + \beta_2$ stays fixed.

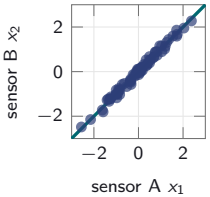
Warning

OLS has no reason to prefer $(2, 2)$ over $(100, -96)$ when the features are exact duplicates. Training fit alone cannot make that choice.

Nearly Duplicate Features: A Large Trade Is Nearly Invisible

Real sensors are rarely identical. Suppose $x_1 \approx x_2$, and again use $v = [1, -1]^\top$. The prediction change is

$$\begin{aligned} \boldsymbol{X}(\beta + tv) - \boldsymbol{X}\beta &= t\boldsymbol{X}v \\ &= t(x_1 - x_2) \\ &\approx 0. \end{aligned}$$



In our six-feature example,

$$\mathrm{corr}(x_1, x_2) = 0.994.$$

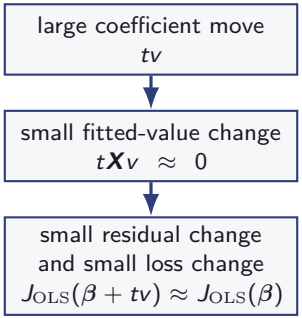
The coefficient trade is no longer exactly invisible, but it is only weakly visible in the predictions.

Large coefficient trade + tiny fitted-value change = a long, nearly flat OLS valley.

Why A Small Prediction Change Creates A Flat OLS Loss

Let $r = \boldsymbol{y} - \boldsymbol{X}\beta$ be the current residual vector. After the coefficient move $\beta \mapsto \beta + tv$, the residual becomes

$$r_{\text{new}} = \boldsymbol{y} - \boldsymbol{X}(\beta + tv) = r - t\boldsymbol{X}v.$$



This is not a new definition.
It follows from the OLS loss we already know:

$$J_{\text{OLS}}(\beta) = \frac{1}{2n}\|r\|_2^2.$$

If $\boldsymbol{X}v$ is tiny, then r_{new} is almost r. Squaring their lengths therefore gives almost the same loss.

At OLS, Only The Quadratic Loss Change Remains

Let $\widehat{\beta}$ be the OLS solution, with residual $\hat{r} = \boldsymbol{y} - \boldsymbol{X}\widehat{\beta}$. Move the coefficients by any vector $\Delta\beta$:

$$\beta_{\text{new}} = \widehat{\beta} + \Delta\beta.$$

Expanding the squared loss gives

$$J_{\text{OLS}}(\widehat{\beta} + \Delta\beta) - J_{\text{OLS}}(\widehat{\beta}) = -\frac{1}{n}\hat{r}^\top \boldsymbol{X}\Delta\beta + \frac{1}{2}\Delta\beta^\top \underbrace{\left(\frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X}\right)}_{G} \Delta\beta.$$

At the OLS minimum, the normal equation says

$$\boldsymbol{X}^\top \hat{r} = 0.$$

Therefore the first term is exactly zero.

So, specifically around the OLS solution,

$$\Delta J = \frac{1}{2}\Delta\beta^\top G \Delta\beta, \qquad G = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X}.$$

Summary

$G = \boldsymbol{X}^\top \boldsymbol{X}/n$ is the OLS Hessian: it measures how quickly the loss rises when we move away from the fitted coefficient vector.

Why Eigenvectors Enter: The OLS Hessian Is Symmetric

The curvature matrix is

$$G = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X} \in \mathbb{R}^{p\times p}.$$

It is symmetric because $G^\top = G$. Therefore the spectral theorem gives

$$\boxed{G = \boldsymbol{Q}\Lambda\boldsymbol{Q}^\top}, \qquad \boldsymbol{Q} = [q_1 \ \cdots \ q_p], \qquad \Lambda = \mathrm{diag}(\mu_1, \ldots, \mu_p).$$

Equivalently, each column q_j satisfies

$$Gq_j = \mu_j q_j, \qquad q_i^\top q_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Meaning of one eigenpair

q_j is a direction in coefficient space. μ_j measures the OLS curvature along that direction. The pair (q_j, μ_j) belongs to the data matrix through $G = \boldsymbol{X}^\top \boldsymbol{X}/n$.

Change Coordinates, Not The Actual Coefficient Move

The same move $\Delta\beta$ can be described in the original axes or in the eigenvector axes:

$$\boxed{z = \boldsymbol{Q}^\top \Delta\beta} \qquad \Longleftrightarrow \qquad \boxed{\Delta\beta = \boldsymbol{Q}z = \sum_{j=1}^{p} z_j q_j.}$$

Here $z_j = q_j^\top \Delta\beta$ is the signed amount of the move along q_j. Substitute the same move into the loss change:

$$\begin{aligned} \Delta J &= \frac{1}{2}(\boldsymbol{Q}z)^\top (\boldsymbol{Q}\Lambda\boldsymbol{Q}^\top)(\boldsymbol{Q}z) \\ &= \frac{1}{2} z^\top \Lambda z = \boxed{\frac{1}{2}\sum_{j=1}^{p} \mu_j z_j^2}. \end{aligned}$$

Insight

No coefficient vector has changed. We only chose coordinates in which the coupled quadratic loss becomes a sum of independent one-direction costs.

A Small Eigenvalue Leaves Many Nearly Equivalent Fits

Suppose we move from OLS along one eigendirection:

$$\beta(t) = \widehat{\beta} + tq, \qquad J_{\mathrm{OLS}}(\beta(t)) = J_{\mathrm{OLS}}(\widehat{\beta}) + \frac{1}{2}\mu t^2.$$

Weak direction: $\mu = 0.002$

t	0	5	10
ΔJ	0	0.025	0.10

Moving ten coefficient units barely changes the loss.

Strong direction: $\mu = 2$

t	0	5	10
ΔJ	0	25	100

The same movement is strongly rejected by the loss.

Warning

Small μ is not *good because the loss stays small*. It means OLS has little loss-based reason to choose one location over another along that direction.

A Flat Loss Makes The Estimated Coefficients Unstable

Dataset 1 has the flat one-dimensional loss

$$J_1(t) = 0.001t^2,$$

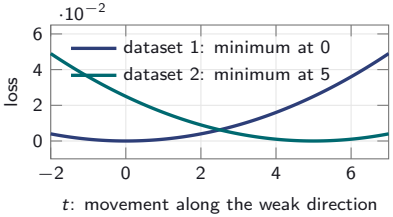
with minimum at $t = 0$. A small data change produces

$$J_2(t) = 0.001t^2 - 0.01t + 0.025 = 0.001(t-5)^2.$$

$$\frac{dJ_2}{dt} = 0.002t - 0.01,$$

$$0.002t - 0.01 = 0,$$

$$\boxed{t_{\text{new}} = 5.}$$



Summary

Low loss curvature means high estimator sensitivity: slightly different training data can produce substantially different OLS coefficients.

The OLS Inverse Reveals The Same Instability

Divide the normal equation by n and define the data-side vector c:

$$G\widehat{\beta}_{\mathrm{OLS}} = c, \qquad G = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X}, \qquad c = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{y}.$$

Using $G = \boldsymbol{Q}\Lambda\boldsymbol{Q}^\top$ and $\boldsymbol{Q}^{-1} = \boldsymbol{Q}^\top$,

$$\boxed{\widehat{\beta}_{\mathrm{OLS}} = G^{-1}c = \boldsymbol{Q}\Lambda^{-1}\boldsymbol{Q}^\top c}, \qquad \Lambda^{-1} = \mathrm{diag}\left(\frac{1}{\mu_1}, \ldots, \frac{1}{\mu_p}\right).$$

eigenvalue	inverse factor
$\mu = 10$	$1/\mu = 0.1$
$\mu = 0.001$	$1/\mu = 1000$

If the sample changes c to $c + \Delta c$, then

$$\Delta\widehat{\beta} = G^{-1}\Delta c.$$

A small component of Δc along q_j is multiplied by $1/\mu_j$.

Warning

If any $\mu_j = 0$, the inverse does not exist and OLS has no unique coefficient solution.

Ridge Prefers Smaller Coefficients When Fits Are Similar

For duplicate features, both coefficient pairs below produce the same prediction because $\beta_1 + \beta_2 = 4$.

$$\boldsymbol{\beta}^{(A)} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}, \qquad \|\boldsymbol{\beta}^{(A)}\|_2^2 = 8. \qquad\qquad \boldsymbol{\beta}^{(B)} = \begin{bmatrix} 10 \\ -6 \end{bmatrix}, \qquad \|\boldsymbol{\beta}^{(B)}\|_2^2 = 136.$$

OLS compares only the residual sum of squares. Ridge adds a second preference:

$$\boxed{J_{\text{ridge}}(\boldsymbol{\beta}) = \frac{1}{2n}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \frac{\lambda}{2}\|\boldsymbol{\beta}\|_2^2.}$$

Insight

Large coefficients are not inherently wrong. Ridge says: among fits with similar training error, prefer the coefficient vector that does not travel unnecessarily far along a weak direction.

The Ridge Normal Equation Adds λI

$$J_{\text{ridge}}(\beta) = \frac{1}{2n}\|\boldsymbol{y} - \boldsymbol{X}\beta\|_2^2 + \frac{\lambda}{2}\beta^\top\beta,$$
$$\nabla_\beta J_{\text{ridge}} = -\frac{1}{n}\boldsymbol{X}^\top(\boldsymbol{y} - \boldsymbol{X}\beta) + \lambda\beta.$$

Set the gradient to zero and rearrange:

$$\frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X}\widehat{\beta}_\lambda + \lambda\widehat{\beta}_\lambda = \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{y},$$
$$\left(\frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X} + \lambda I\right)\widehat{\beta}_\lambda = \frac{1}{n}\boldsymbol{X}^\top\boldsymbol{y}.$$

Here $\lambda\beta = \lambda I\beta$ because $I\beta = \beta$. Therefore

$$\boxed{\widehat{\beta}_\lambda = \left(\frac{1}{n}\boldsymbol{X}^\top\boldsymbol{X} + \lambda I\right)^{-1}\frac{1}{n}\boldsymbol{X}^\top\boldsymbol{y}.}$$

Ridge Adds λ To Every Eigenvalue

Let

$$\boldsymbol{G} = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X}, \qquad \boldsymbol{G}q_j = \mu_j q_j.$$

Apply the Ridge matrix to the same eigenvector:

$$\begin{aligned}(\boldsymbol{G} + \lambda I)q_j &= \boldsymbol{G}q_j + \lambda I q_j \\ &= \mu_j q_j + \lambda q_j \\ &= \boxed{(\mu_j + \lambda)q_j}.\end{aligned}$$

Weak OLS direction

$\mu_j = 0.001$, $\lambda = 1$:

$$\frac{1}{\mu_j} = 1000 \quad \longrightarrow \quad \frac{1}{\mu_j + \lambda} \approx 0.999.$$

Strong OLS direction

$\mu_j = 100$, $\lambda = 1$:

$$\frac{1}{\mu_j} = 0.010 \quad \longrightarrow \quad \frac{1}{\mu_j + \lambda} \approx 0.0099.$$

Project The Normal Equations Onto One Eigenvector

For one eigenvector q_j, define

$$a_j = q_j^\top \beta, \qquad c_j = q_j^\top c, \qquad c = \frac{1}{n} X^\top y.$$

Since $q_j^\top G = \mu_j q_j^\top$, multiply each normal equation by $q_j^\top$:

	project onto q_j	**solve the scalar equation**
OLS: $G\widehat{\beta} = c$	$\mu_j a_j^{\mathrm{OLS}} = c_j$	$a_j^{\mathrm{OLS}} = \dfrac{c_j}{\mu_j}$
Ridge: $(G + \lambda I)\widehat{\beta} = c$	$(\mu_j + \lambda) a_j^{\mathrm{ridge}} = c_j$	$a_j^{\mathrm{ridge}} = \dfrac{c_j}{\mu_j + \lambda}$

Divide the Ridge coordinate by the OLS coordinate:

$$\boxed{a_j^{\mathrm{ridge}} = \frac{\mu_j}{\mu_j + \lambda}\, a_j^{\mathrm{OLS}}.}$$

Large μ_j: factor near 1. Small μ_j: factor near 0.

One Dot Is One Model Fitted To A Full Dataset

Repeat $r = 1, \ldots, 60$
new dataset D_r
42 observations

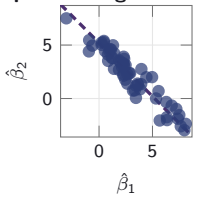
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fit both models
to all 42 rows
OLS and Ridge

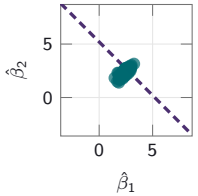
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record two estimates
one OLS dot
one Ridge dot

60 OLS fits: spread along the weak direction



60 Ridge fits: estimates cluster more tightly



Each panel has 60 dots. Every dot comes from a full 42-observation dataset.

Bias And Variance Ask Two Different Questions

Suppose the true coefficient in one direction is $\beta^\star = 2$. Refit the model on many new training samples.

OLS estimates

$$-1, \quad 5, \quad 0, \quad 4, \quad 2$$

Average $= 2$: little bias.
The estimates are widely spread: high variance.

Ridge estimates

$$1.2, \quad 1.6, \quad 1.3, \quad 1.5, \quad 1.4$$

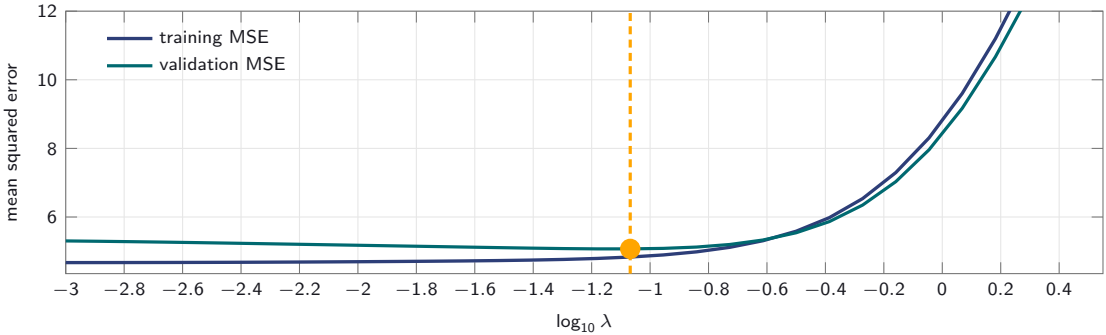
Average $= 1.4$: some bias toward zero.
The estimates are tightly clustered: lower variance.

$$\boxed{\text{expected test error} = \text{bias}^2 + \text{variance} + \text{irreducible noise.}}$$

Summary

Ridge can improve prediction when a modest increase in bias buys a much larger reduction in sampling variance.

Validation Chooses The Regularization Strength



Summary

Training error always prefers $\lambda = 0$ because OLS already minimizes it. Validation error can select a positive λ that predicts new cases better.

Lasso Changes Only The Penalty

The fit term is unchanged. Lasso replaces squared coefficient length by total absolute coefficient size.

Ridge

$$\min_{\beta} \frac{1}{2n}\|y - X\beta\|_2^2 + \frac{\lambda}{2}\sum_{j=1}^{p}\beta_j^2.$$

Smooth shrinkage; coefficients usually remain nonzero.

Lasso

$$\min_{\beta} \frac{1}{2n}\|y - X\beta\|_2^2 + \lambda\sum_{j=1}^{p}|\beta_j|.$$

Shrinkage with the possibility of exact zeros.

Why can replacing β_j^2 by $|\beta_j|$ create an exact zero?

Summary

Two answers: the ℓ_1 budget has axis corners, and the one-coordinate solution has a threshold at zero.

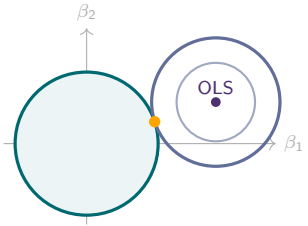
The Lasso Diamond Makes Exact Zeros Possible

For a corresponding budget s, the penalized solution can also be viewed as minimizing OLS loss subject to a coefficient budget:

$$\text{Ridge: } \|\beta\|_2 \leq s, \qquad \text{Lasso: } \|\beta\|_1 \leq s.$$

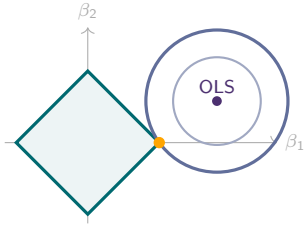
Larger λ means a stronger penalty and corresponds to a smaller budget s.

Ridge: smooth ℓ_2 circle



Smooth boundary: contact is generally off the axes.

Lasso: ℓ_1 diamond with axis corners



Axis corner: one coefficient can be exactly zero.

Summary

The ℓ_1 corners lie on coordinate axes. Lasso can hit a corner, although it does not always do so.

Where Do b And z In Soft Thresholding Come From?

For this derivation, assume standardized orthogonal columns and define

$$G = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{X} = I, \qquad z = \frac{1}{n}\boldsymbol{X}^\top \boldsymbol{y}.$$

Then $G\widehat{\boldsymbol{\beta}}_{\mathrm{OLS}} = z$ implies $z = \widehat{\boldsymbol{\beta}}_{\mathrm{OLS}}$. Expand the full Lasso objective and complete the square in the last line:

$$\begin{aligned} J_{\mathrm{lasso}}(\boldsymbol{\beta}) &= \frac{1}{2n}\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j| \\ &= \text{constant} + \frac{1}{2}\boldsymbol{\beta}^\top G\boldsymbol{\beta} - z^\top \boldsymbol{\beta} + \lambda \sum_{j=1}^{p} |\beta_j| \\ &= \text{constant} + \sum_{j=1}^{p} \left[\frac{1}{2}(\beta_j - z_j)^2 + \lambda|\beta_j|\right]. \end{aligned}$$

The sum separates. For coordinate j, rename

$$\boxed{b = \beta_j, \qquad z = z_j = \hat{\beta}_{j,\mathrm{OLS}}} \quad \Longrightarrow \quad \min_b \ \frac{1}{2}(b - z)^2 + \lambda|b|.$$

This one-variable problem is one coordinate of the full Lasso objective.

Soft Thresholding Explains Exact Zeros

For one standardized, orthogonal coordinate, Lasso solves

$$\min_b \ \frac{1}{2}(b-z)^2 + \lambda|b|,$$

where z is the coordinate's OLS estimate.

$z > \lambda$

If $b > 0$, then $|b| = b$:

$$(b-z) + \lambda = 0,$$
$$b = \boxed{z - \lambda > 0.}$$

$z < -\lambda$

If $b < 0$, then $|b| = -b$:

$$(b-z) - \lambda = 0,$$
$$b = \boxed{z + \lambda < 0.}$$

$|z| \le \lambda$

Neither nonzero case is valid. The optimum is at the kink:

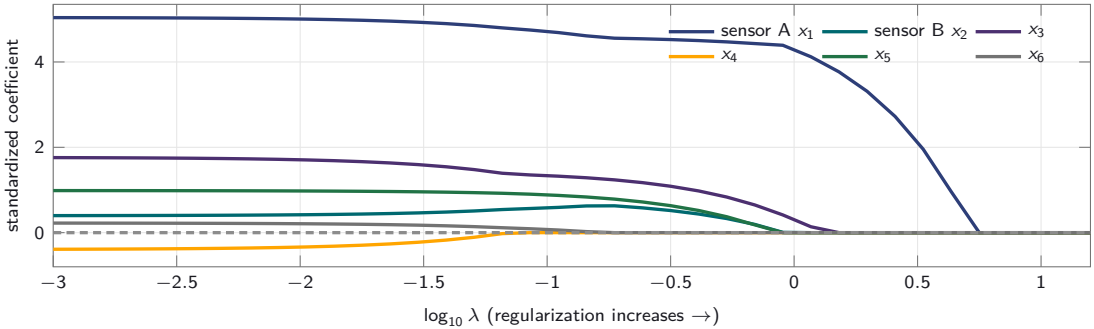
$$\boxed{b = 0.}$$

$$\boxed{\hat{b}_{\text{lasso}} = \text{sign}(z)\max(|z| - \lambda, 0).}$$

Summary

Ridge smoothly rescales a signal. Lasso thresholds it, so a small OLS coordinate becomes exactly zero.

Lasso Creates A Sequence Of Increasingly Sparse Models



Summary

As λ grows, weaker paths reach zero and stay there. Each value of λ defines a different candidate feature subset.

Standardize Before Penalizing Coefficients

Suppose the same rainfall predictor is recorded in metres or millimetres:

$$x_{\mathrm{mm}} = 1000 x_{\mathrm{m}}, \qquad \beta_{\mathrm{mm}} = \frac{\beta_{\mathrm{m}}}{1000}.$$

The prediction is unchanged, but the squared penalty changes:

$$\beta_{\mathrm{mm}}^2 = 10^{-6} \beta_{\mathrm{m}}^2.$$

Warning

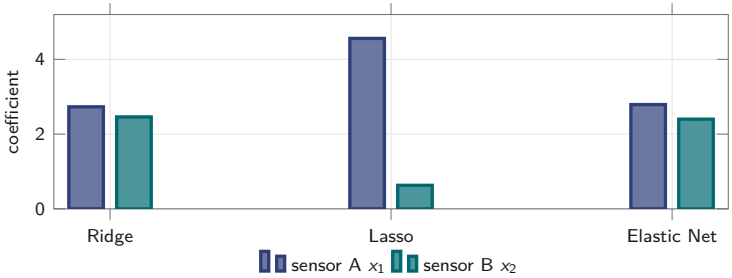
Without standardization, Ridge and Lasso penalize measurement units rather than comparable coefficient magnitudes.

$$\widetilde{x}_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j} \quad \Longrightarrow \quad \text{one coefficient unit means one predictor standard deviation.}$$

Center the response and handle the intercept separately. The intercept is normally left unpenalized.

Lasso May Choose One Correlated Predictor

The two sensor columns contain almost the same information. Their selected coefficients are:

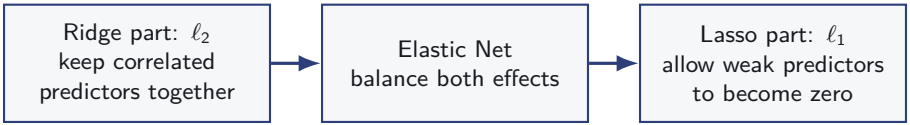


Warning

Do not interpret Lasso's larger sensor-A coefficient as evidence that sensor A causes more flooding. Under strong collinearity, small data changes can alter which representative receives the larger coefficient.

Elastic Net Uses Two Complementary Forces

Elastic Net is useful when predictors are correlated but some predictors may still be irrelevant.



$$\hat{\boldsymbol{\beta}}^{\mathrm{EN}}_{\lambda,\alpha} = \arg\min_{\boldsymbol{\beta}} \left\{ \frac{1}{2n} \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|_2^2 + \lambda \left[\alpha \|\boldsymbol{\beta}\|_1 + \frac{1-\alpha}{2} \|\boldsymbol{\beta}\|_2^2 \right] \right\}.$$

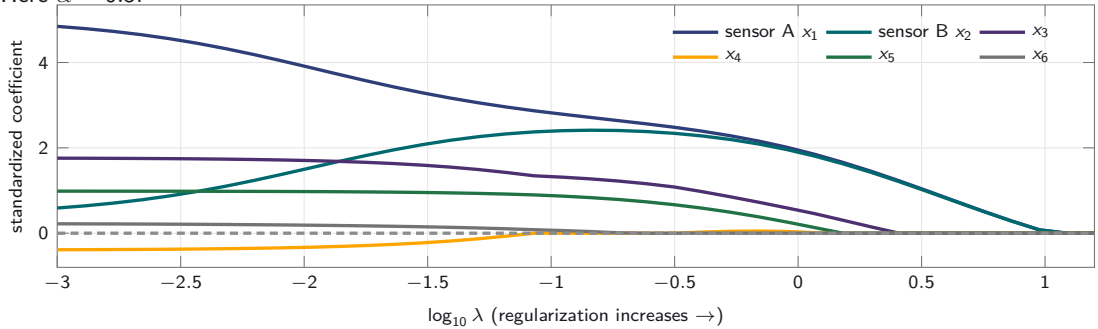
$\alpha = 0$	$0 < \alpha < 1$	$\alpha = 1$
Ridge.	Stable sparsity for correlated groups.	Lasso.

Insight

The ℓ_2 term shares effect across correlated sensors; the ℓ_1 term can still remove unsupported predictors.

Elastic Net Keeps Correlated Paths Together

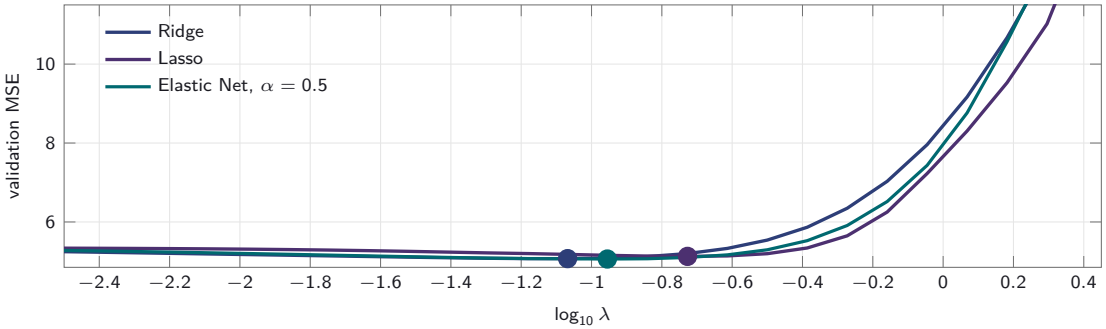
Here $\alpha = 0.5$.



Summary

The sensor paths approach one another as regularization grows, while weak predictors can still reach zero.

Validation Chooses A Method And Its Regularization Strength



Summary

No method wins by definition. On this run Elastic Net has the lowest validation MSE, but the three minima are close enough that uncertainty should be reported.

What Each Method Assumes About A Useful Model

	Ridge	Lasso	Elastic Net
Penalty	$\|\beta\|_2^2$	$\|\beta\|_1$	$\ell_1 + \ell_2$
Coefficient behavior	smooth shrinkage; usually no zeros	some exact zeros	zeros plus group shrinkage
Natural use case	many small or correlated effects	a genuinely sparse signal	sparse signal with correlated groups
Main caution	all variables remain	selected variable can change under collinearity	two tuning choices, λ and α

Summary

The choice is a modeling hypothesis, not a ranking of fashionable algorithms. Use validation performance, stability, and the scientific purpose together.

Regularization Also Extends To Logistic Regression

Only the data-fit term changes.

$$\widehat{\beta} = \arg\min_{\beta} \left\{ \underbrace{-\sum_{i=1}^{n} [y_i \log p_i + (1 - y_i) \log(1 - p_i)]}_{\text{logistic negative log-likelihood}} + \lambda R(\beta) \right\},$$

where

$$p_i = \sigma(x_i^\top \beta).$$

$R(\beta) = \|\beta\|_2^2$ gives logistic Ridge.

$R(\beta) = \|\beta\|_1$ gives logistic Lasso.

Summary

Regularization is a general modification of an objective function, not a trick limited to the closed-form linear regression solution.

A Professional Regularization Workflow

1. Split the observations into training and final test data.
2. Within the training data, center and standardize predictors without using information from validation folds.
3. Choose a grid of λ values; for Elastic Net, also choose a small grid of α values.
4. Use cross-validation to estimate future error for each setting.
5. Select a setting, refit on all training data, and evaluate once on the untouched test data.
6. Report predictive performance, selected variables, coefficient stability, and the complete preprocessing procedure.

Warning

Never standardize the full dataset before splitting, and never use the test set to tune λ.

What Students Should Now Be Able To Explain

1. Correlated predictors create weak eigen-directions in $X^\top X$ and make OLS coefficients unstable.
2. Ridge adds λ to each eigenvalue and shrinks weak directions most.
3. Validation selects the bias-variance compromise; training error cannot.
4. Lasso's ℓ_1 kink creates soft thresholding and exact zeros.
5. Elastic Net can preserve correlated groups while retaining sparsity.
6. The same penalty idea applies to likelihood-based models such as logistic regression.

Exit check

For two nearly identical predictors, explain why Ridge tends to share their effect while Lasso may choose one representative.

Pedagogical reference: James, Witten, Hastie, Tibshirani, and Taylor, *An Introduction to Statistical Learning*, 2nd ed., Chapter 6. All plotted results in this deck are regenerated from the accompanying seeded simulation.